

EXP.NO: 3

Write a Servlet program that prints "Hello, World!" when accessed through a browser.

AIM:

To write a basic Servlet program that demonstrates the use of Java Servlets by printing "Hello, World!" when accessed through a browser.

ALGORITHM:

1. Input:

- The user makes an HTTP request (usually by entering a URL) through a browser.

2. Servlet Program Flow:

1. Create a Servlet Class:

- Define a Java class that extends the `HttpServlet` class.
- Override the `doGet()` method, which handles HTTP GET requests from the client (browser).

2. Handle HTTP Request:

- In the `doGet()` method:
 - Get the **PrintWriter** object to write the response to the browser.
 - Set the response content type to **text/html**.
 - Write the **"Hello, World!"** message to the response.

3. Compile the Servlet Class:

- Compile the Java Servlet file using the `javac` command to generate a `.class` file.

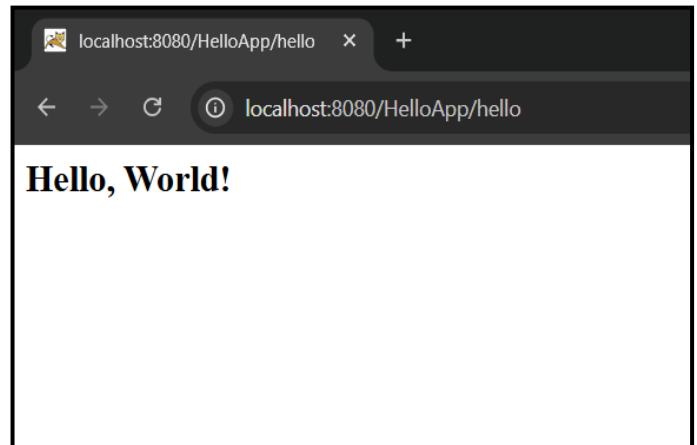
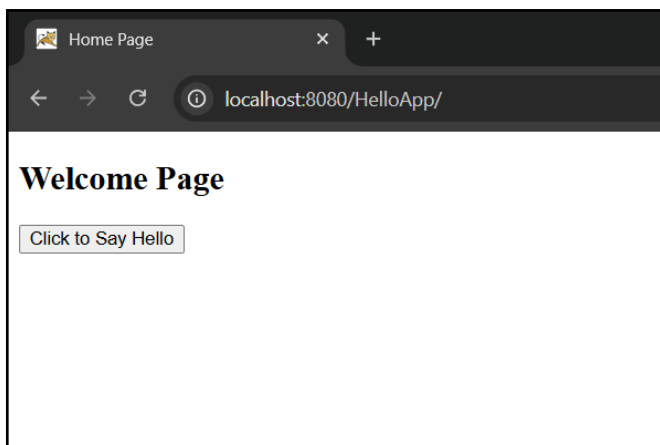
4. Deploy the Servlet:

- Place the compiled servlet class file in the appropriate directory of the web application (typically in the `/WEB-INF/classes` folder).
- Configure the servlet in the `web.xml` file of the web application or use annotations.

5. Access the Servlet via Browser:

- Start the **Apache Tomcat** or another servlet container.
- Access the servlet by entering the appropriate URL (e.g., `http://localhost:8080/your-webapp/hello`).

Output:



Java File (HelloServlet.java) :

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.*;
import jakarta.servlet.http.*;

public class HelloServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        out.println("<html><body>");
        out.println("<h2>Hello, World!</h2>");
        out.println("</body></html>");

    }
}
```

File (web.xml) :

```
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
    version="5.0">

    <welcome-file-list>
        <welcome-file>index.html</welcome-file>
    </welcome-file-list>

    <servlet>
        <servlet-name>HelloServlet</servlet-name>
        <servlet-class>HelloServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>HelloServlet</servlet-name>
        <url-pattern>/hello</url-pattern>
    </servlet-mapping>

</web-app>
```

HTML File (index.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>Home Page</title>
</head>
<body>

  <h2>Welcome Page</h2>

  <a href="hello">
    <button>Click to Say Hello</button>
  </a>

</body>
</html>
```

RESULT:

The browser will display the message "Hello, World!" returned by the Servlet.